**Project Design Phase**
**Problem – Solution Fit**

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| Date | 3 February 2026 |
| Team ID | LTVIP2026TMIDS79347 |
| Project Name | Online payment fraud detection using machine learning |
| Maximum Marks | 2 Marks |

**Problem – Solution Fit**

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| **1. Customer Segments (CS)**<br>Online payment users, e-commerce businesses, banks, and fintech companies.<br>Organizations that process large volumes of digital transactions. | **6. Customer Constraints (CC)**<br>Large transaction volumes and evolving fraud techniques.<br>Risk of false positives affecting genuine users. | **5. Available Solutions (AS)**<br>OTP verification and rule-based fraud systems.<br>Manual fraud review teams with limited adaptability. |
| **2. Jobs-To-Be-Done / Problems (JBP)**<br>Ensure secure and safe online transactions.<br>Detect and prevent fraudulent activities in real time. | **9. Problem Root Cause (RC)**<br>Weak authentication and static rule-based systems.<br>Lack of behavioral pattern analysis and delayed detection. | **7. Behaviour (BE)**<br>Frequent online transactions and changing spending patterns.<br>Fraudsters exploit system weaknesses and stolen credentials. |
| **3. Triggers (TR)**<br>High-value transactions or unusual login locations.<br>Multiple failed login attempts or sudden spending changes.<br><br>**4. Emotions: Before / After (EM)**<br>**Before:** Fear of fraud and financial loss.<br>**After:** Confidence, trust, and secure payment experience. | **10. Your Solution (SL)**<br>A machine learning–based system that detects fraud in real time.<br>Continuously learns from transaction data to improve accuracy. | **8. Channels of Behaviour (CH)**<br>Web applications, payment gateways, and banking systems.<br>SMS/Email alerts and customer support verification. |